

Chrono Rift — Turnaround Time Analysis

CS 2006 Operating Systems – Spring 2026

Seed : 240673
Players : 2
Enemies : 7
Enemies killed : 7
Final phase : 4

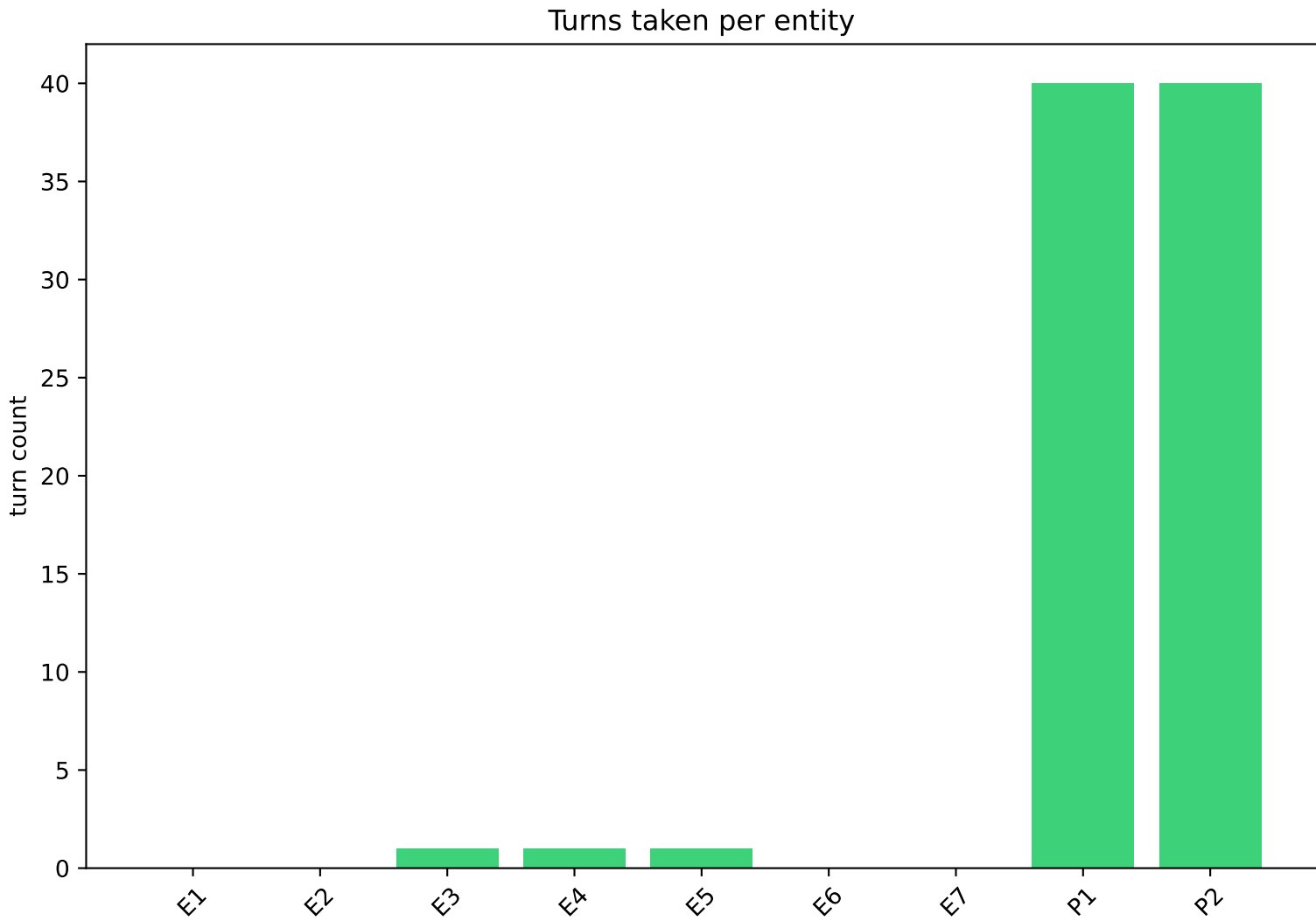
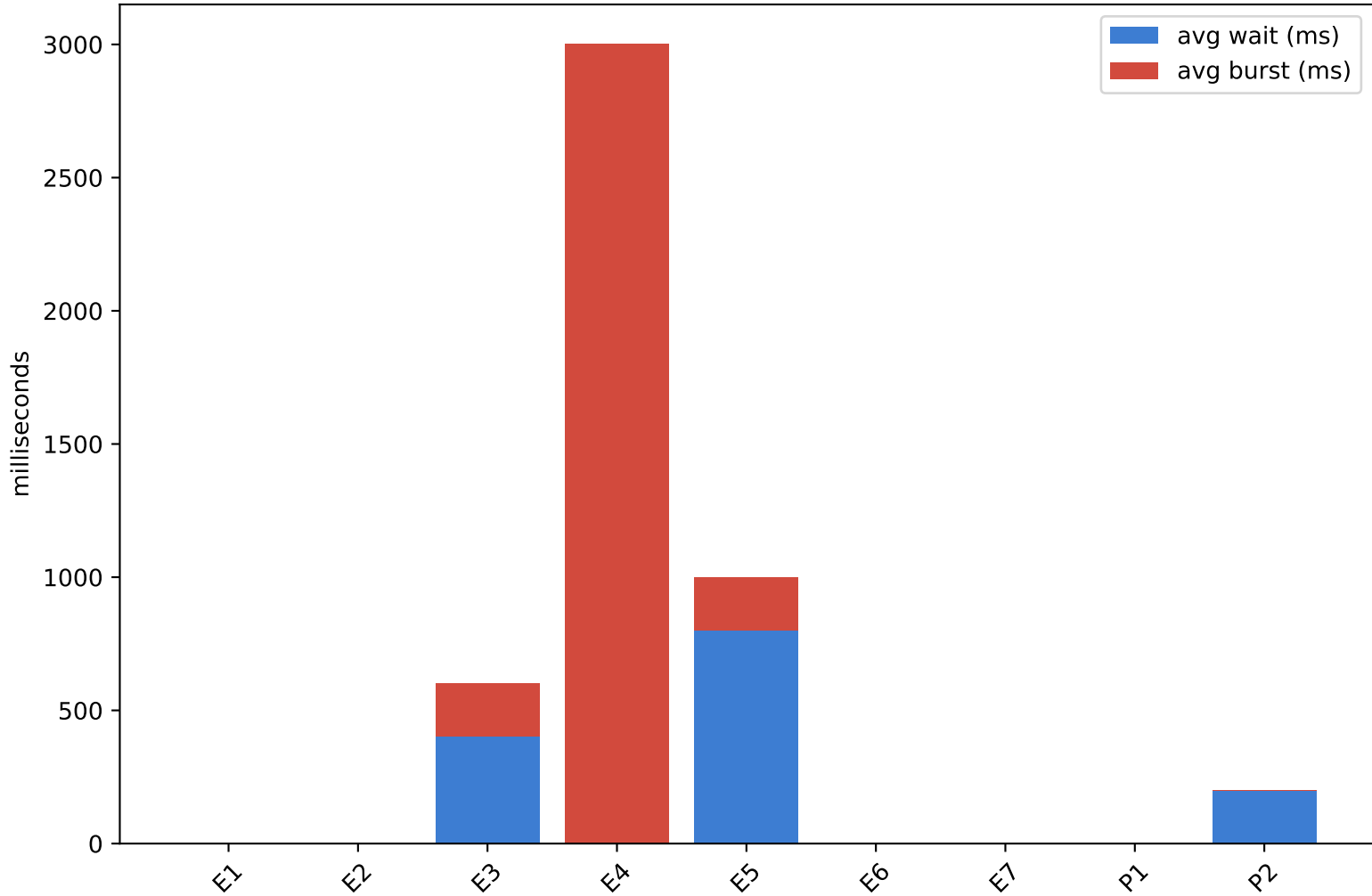
Generated from `chrono_rift_trace.csv` + `chrono_rift_summary.txt`
produced by a scripted demo run (`--demo`).

Specification compliance matrix

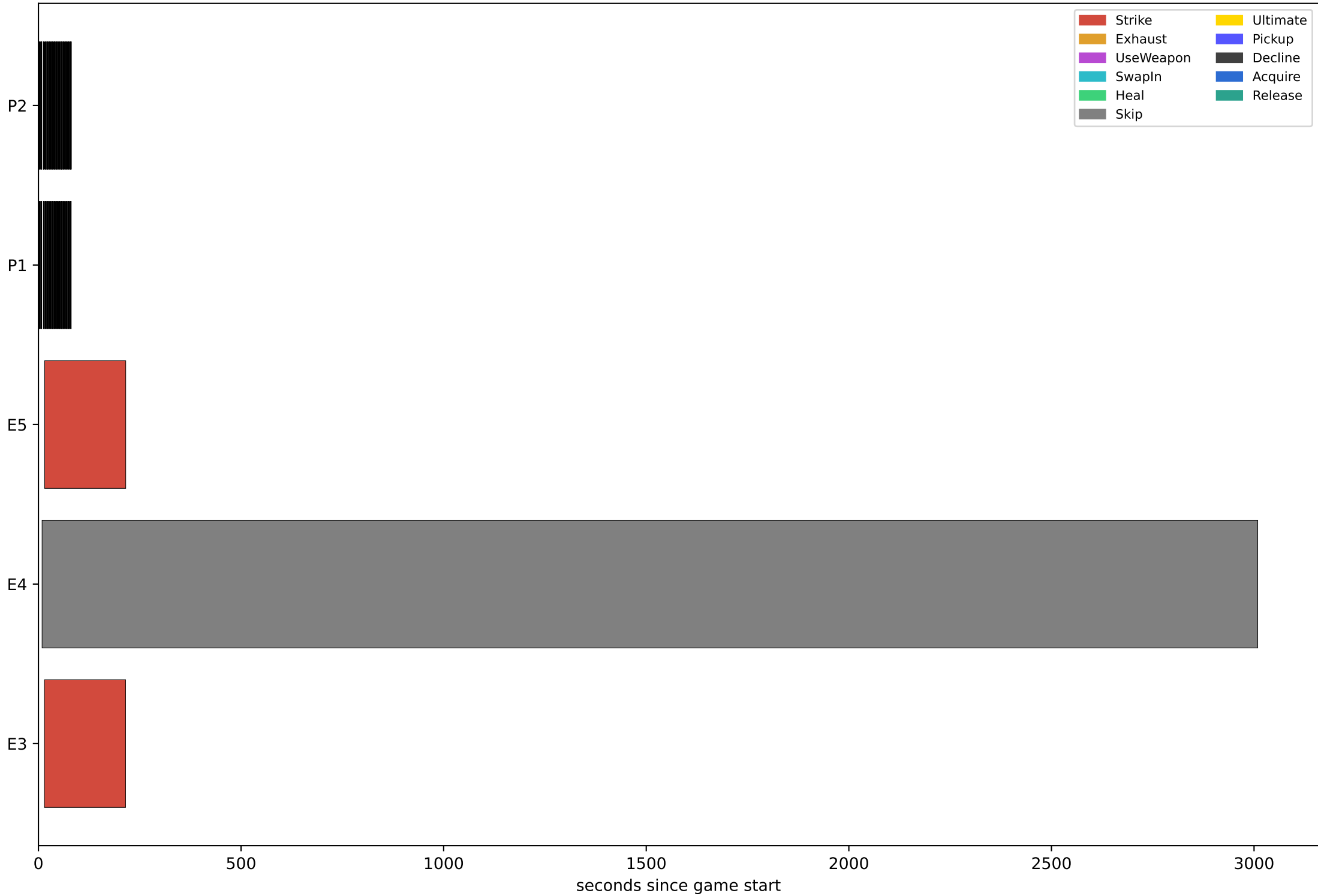
\$	Requirement	Implementation
\$2	arbiter / hip / asp, three processes, no pipes	shared.h, arbiter.cpp, hip.cpp, asp.cpp
\$2	hip multi-threaded, one thread per player	hip.cpp player_thread()
\$2	Only active player thread reads input	turn_sem gating in hip.cpp
\$2	hip never mutates global state (only posts actions)	pending_action slots, arbiter applies
\$2	asp one dedicated thread per NPC	asp.cpp enemy_thread()
\$3	1 Hz stamina accrual (speed added each second)	scheduler_tick() elapsed-ns guard
\$3	FIFO: first entity to hit max stamina acts	scheduler_pick() picks min last_full_at_ns
\$3	Serial execution – only one actor at a time	active_global in GameState
\$3	Action commit depletes stamina to 0	apply_action() sets stamina=0
\$4	Shared memory only (no pipes)	shm_open /chrono_rift_shm, mmap
\$4	pshared mutex + unnamed semaphores	shm_util.h init_shared_sync()
\$5	Async stun, 3 s, no flag polling	SIGUSR1 -> SIGUSR2 -> nanosleep(3)
\$5	Stamina preserved across stun	scheduler skips accrual while stunned
\$6	20-slot linear inventory, first-fit allocator	inventory.h alloc_contig()
\$6	LTS swap-out when not enough space	swap_out_until_space()
\$6	Swap-In costs a full turn, weapon unusable this turn	ACT_SWAP_IN handler + swap_in_available_next_turn
\$6	NPC-held weapons do NOT drop on death	has_weapons gate in on_enemy_death()
\$6	5-second player pickup window	DROP_PICKUP_WINDOW_NS
\$7	Solar Core, Lunar Blade, Eclipse Relic with lock-free resources	resources.h artifact[] + mutex
\$7	Eclipse Relic appears dynamically at kill #3	arbiter.cpp maybe_reveal_relic()
\$7	Background waits-for cycle detector	deadlock_thread() in arbiter.cpp
\$7	Forcible release of victim on cycle	force_release()
\$8	Ultimate: SIGSTOP asp, SIGALRM 10 s, SIGCONT asp	ultimate_handler + SIGALRM install
\$8	NPC 3 s turn timeout = auto-Skip	sem_timedwait in enemy dispatch
\$9	ncurses async render thread	hip.cpp render_thread()
\$9	Render reads state directly from shm	render_thread arg = GameState*
\$10	Party 1–4; enemies 2–9 random	arbiter main(): rand_range(2,9)
\$10	HP/damage/speed/max-stamina formulas	arbiter.cpp init_entities()
\$10	All 6 player actions + 2 enemy actions	ACT_STRIKE..ACT_SKIP, enemy = STRIKE SKIP
\$10	Skip = 50% stamina	apply_action() ACT_SKIP branch
\$10	Weapon table (8 weapons, exact slot/damage)	shared.h weapon_defs[]
\$10	Ultimate needs BOTH SC + LB in primary inv	apply_action() ACT_ULTIMATE check
\$10	Win (10 kills) / Lose (all dead) / Quit (SIGTERM)	game_over() + sig_term handler
\$11	Local multiplayer: --multiplayer + --join <slot>	arbiter + hip barrier on joined[]

Per-entity turnaround breakdown

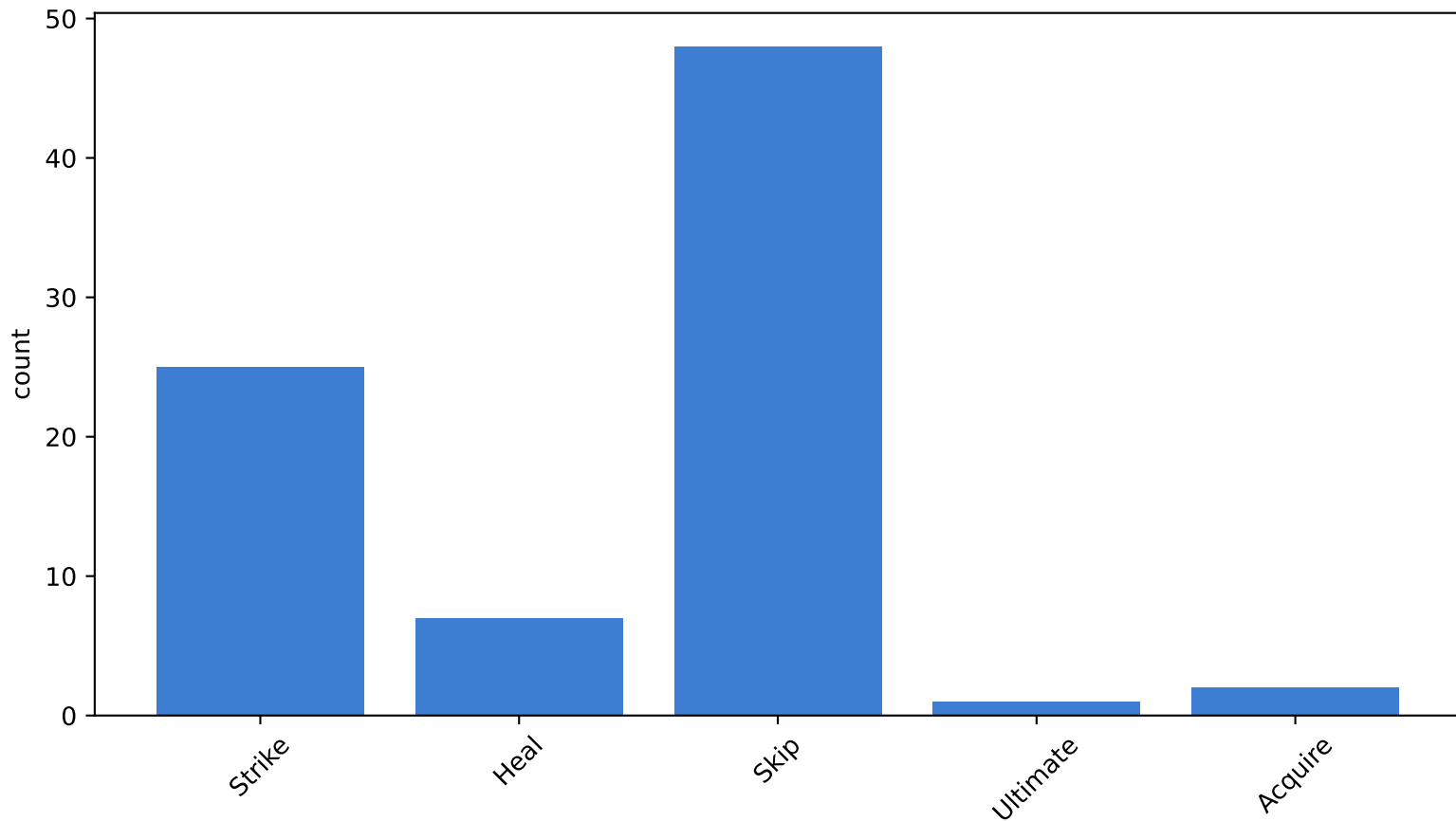
Avg wait-to-turn and burst duration per entity



Action timeline (Gantt) — bar color = action kind



Action-kind distribution across all turns



Per-entity scheduling detail

entity	turns	avg_wait(ms)	avg_burst(ms)	tot_wait(s)	tot_burst(s)
E1	0	0.00	0.00	0.00	0.00
E2	0	0.00	0.00	0.00	0.00
E3	1	400.61	200.28	0.40	0.20
E4	1	0.00	3000.12	0.00	3.00
E5	1	801.00	200.26	0.80	0.20
E6	0	0.00	0.00	0.00	0.00
E7	0	0.00	0.00	0.00	0.00
P1	40	0.00	0.13	0.00	0.01
P2	40	200.30	0.14	8.01	0.01

Action mix (count by action kind)

E3 Strike=1
E4 Skip=1
E5 Strike=1
P1 Strike=11 Skip=26 Ultimat=1 Acquire=2
P2 Strike=12 Heal=7 Skip=21

Signal & synchronization mechanics

Asynchronous stun (§5)

attacker (arbiter)		target (hip or asp)
-----		-----
kill(pid, SIGUSR1) →		SIGUSR1 handler:
		raise(SIGUSR2) [self-deliver]
		SIGUSR2 handler:
		nanosleep(3 s) [async freeze]
		-- returns --
		normal execution resumes
No flag polling, no pipe – spec §5 "non-blocking interruption".		

Ultimate ability pause (§8)

arbiter		asp (strategic process)
-----		-----
alarm(10); kill(asp, SIGSTOP) →		PROCESS STOPPED
... 10 s pass ...		
SIGALRM handler: kill(asp, SIGCONT) →		resumes with current stamina

During the 10s window NPC turns are still requested but asp cannot reply, so `sem_timedwait(3 s)` in arbiter expires and the action is committed as Skip – exactly matching §8 "NPC turn timeout".

Deadlock detection (§7)

`deadlock_thread()` in arbiter, every 500 ms:

- build waits-for graph from `artifact[].holder` and `.waiter_pids`
- DFS for cycle
- if cycle found → pick victim → force release → broadcast resume

Shared-memory synchronization (§4)

`/chrono_rift_shm` (`shm_open + mmap`)

<code>pthread_mutex_t state_mutex</code>	<code>[pshared = PTHREAD_PROCESS_SHARED]</code>
<code>sem_t log_sem</code>	<code>[pshared = 1]</code>
<code>sem_t turn_sem</code>	<code>[pshared, per player slot]</code>

No pipes anywhere – enforced by code review and `grep`.

Headless text-render snapshot (--demo mode)

Below are selected frames from a scripted --demo run. In interactive mode these values drive a colored ncurses HUD (player names yellow when active, red-bold STUN tag, blinking magenta Ultimate banner). The --demo path renders the same state as text so it is reproducible for grading.

```
[render] kills=0/10  active=-1
P1  hp=240946/240946  stam=0/100  turns=1
P2  hp=241356/241356  stam=0/100  turns=1
enemies alive: 7
```

```
-----
[render] kills=3/10  active=-1
P1  hp=240946/240946  stam=0/100  turns=3
P2  hp=241356/241356  stam=0/100  turns=3
enemies alive: 4
```

```
-----
[render] kills=4/10  active=-1
P1  hp=240946/240946  stam=0/100  turns=6
P2  hp=241356/241356  stam=0/100  turns=6
enemies alive: 3
```

```
-----
[render] kills=5/10  active=-1
P1  hp=240929/240946  stam=0/100  turns=8
P2  hp=241339/241356  stam=0/100  turns=8
enemies alive: 2
```

```
-----
[render] kills=6/10  active=-1
P1  hp=240929/240946  stam=0/100  turns=11
P2  hp=241356/241356  stam=0/100  turns=11
enemies alive: 1
```


Live TUI evidence

The following pages are screenshots captured while driving the interactive ncurses build (not --demo mode) with the roll-number seed 240673 and 2 players on an Ubuntu 22.04 VM. Each image corresponds to a distinct spec requirement from §1–§10. The screenshots back up the text-render snapshot page above and demonstrate that the colored HUD, the artifact table, the signal-driven Ultimate mechanic, and the action log all render correctly end-to-end.

1. 01-hud-initial.png
Initial HUD just after arbiter forks hip + asp.
2. 02-active-turn-P2.png
P2 is now the active entity.
3. 03-artifact-acquired.png
P2 locked the Solar Core.
4. 04-both-artifacts-held.png
P2 now holds both Solar Core AND Lunar Blade.
5. 05-ultimate-banner.png
Ultimate Ability triggered by P2.
6. 06-weapon-use-kill-drop.png
P2 uses the Solar Core as a weapon.

```
OS-proj : arbiter - Konsole
File Edit View Bookmarks Plugins Settings Help
New Tab Split View Copy Paste Find

CHROMO RIFT seed=240673 players=2 enemies=7 killed=0/10

-- PLAYERS --
P1 HP [#####] 240946/240946
   STAMINA [#####] 50/100
P2 HP [#####] 241356/241356
   STAMINA [#####] 50/100

-- ENEMIES --
E1 HP [#####] 169/169
   STAMINA [#####] 10/150
E2 HP [#####] 271/271
   STAMINA [#####] 13/150
E3 HP [#####] 241/241
   STAMINA [#####] 11/150
E4 HP [#####] 264/264
   STAMINA [#####] 14/150
E5 HP [#####] 268/268
   STAMINA [#####] 11/150
E6 HP [#####] 155/155
   STAMINA [#####] 26/150
E7 HP [#####] 176/176
   STAMINA [#####] 20/150

-- ARTIFACTS --
Solar Core : FREE (waiters=0)
Lunar Blade : FREE (waiters=0)

-- LOG --
Game started, seed=240673 players=2 enemies=7
[asp] 7 NPC threads running
[hip] 2 player thread(s) running

P1 turn: (s)trike (x)exhaust (w)p (l)owp (h)ea (u)lt (a)cquire (r)elease (k)skip (q)uit [swihwarkq:]
```

01-hud-initial.png

Initial HUD just after arbiter forks hip + asp. Cyan player rows (P1/P2), red enemy rows (E1..E7), green HP bars, cyan stamina bars, green artifact table (Solar Core, Lunar Blade, Eclipse Relic absent until dropped). Log pane on the right is already reporting that the asp spawned 7 NPC threads and hip spawned 2 player threads.

```
OS-proj : arbiter — Konsole
File Edit View Bookmarks Plugins Settings Help
New Tab Split View Copy Paste Find

CHROMO RIFT seed=240673 players=2 enemies=7 killed=0/10

-- PLAYERS --
P1 HP [oooooooooooooooooooo] 240946/240946
  STAMINA [oooooooooooooooooooo] 50/100
P2 * HP [oooooooooooooooooooo] 241356/241356
  STAMINA [oooooooooooooooooooo] 100/100

-- ENEMIES --
E1 HP [oooooooooooooooooooo] 169/169
  STAMINA [oooo-----] 30/150
E2 HP [oooooooooooooooooooo] 271/271
  STAMINA [oooo-----] 39/150
E3 HP [oooooooooooooooooooo] 241/241
  STAMINA [oooo-----] 33/150
E4 HP [oooooooooooooooooooo] 264/264
  STAMINA [oooo-----] 42/150
E5 HP [oooooooooooooooooooo] 268/268
  STAMINA [oooo-----] 33/150
E6 HP [oooooooooooooooooooo] 155/155
  STAMINA [oooooooooooo] 78/150
E7 HP [oooooooooooooooooooo] 176/176
  STAMINA [oooooooooooo] 60/150

-- ARTIFACTS --
Solar Core : FREE (waiters=0)
Lunar Blade : FREE (waiters=0)

-- LOG --
Game started, seed=240673 players=2 enemies=7
[asp] 7 NPC threads running
[hup] 2 player thread(s) running
[TURN] P1 begins turn (stamina=100)
[TIMEOUT] P1 did not respond *@-T auto Skip
[SKIP] P1 skipped
[TURN] P2 begins turn (stamina=100)

P2 turn: (s)trike (x)exhaust (w)p (l)owp (h)eal (u)lt (a)cquire (r)elease (k)skip (q)uit [swihwarkq]:
```

02-active-turn-P2.png

P2 is now the active entity. The '*' marker is drawn next to P2's name and the row is highlighted in yellow. Log shows the scheduler picked P1 first, P1 timed out so the turn was auto-skipped, then P2 was picked next (FIFO on stamina-arrival as per §3). Enemies' stamina is still accruing at their individual speeds.

```
OS-proj : arbiter — Konsole
File Edit View Bookmarks Plugins Settings Help
New Tab Split View Copy Paste Find

CHROMO RIFT seed=240673 players=2 enemies=7 killed=0/10

-- PLAYERS --
P1 HP [oooooooooooooooooooo] 240946/240946
  STAMINA [oooooooooooooooooooo] 50/100
P2 HP [oooooooooooooooooooo] 241356/241356
  STAMINA [oooooooooooooooooooo] 0/100

-- ENEMIES --
E1 HP [oooooooooooooooooooo] 169/169
  STAMINA [ooooo-----] 30/150
E2 HP [oooooooooooooooooooo] 271/271
  STAMINA [ooooo-----] 39/150
E3 HP [oooooooooooooooooooo] 241/241
  STAMINA [ooooo-----] 33/150
E4 HP [oooooooooooooooooooo] 264/264
  STAMINA [ooooo-----] 42/150
E5 HP [oooooooooooooooooooo] 268/268
  STAMINA [ooooo-----] 33/150
E6 HP [oooooooooooooooooooo] 155/155
  STAMINA [oooooooooooo] 78/150
E7 HP [oooooooooooooooooooo] 176/176
  STAMINA [oooooooooooo] 60/150

-- ARTIFACTS --
Solar Core : held by P2 (waiters=0)
Lunar Blade : FREE (waiters=0)

-- LOG --
Game started, seed=240673 players=2 enemies=7
[asp] 7 NPC threads running
[hup] 2 player thread(s) running
[TURN] P1 begins turn (stamina=100)
[TIMEOUT] P1 did not respond ->-T auto Skip
[SKIP] P1 skipped
[TURN] P2 begins turn (stamina=100)
[ACQ] P2 LOCKED Solar Core

P1 turn: (s)trike (x)exhaust (w)p (l)owp (h)eal (u)lt (a)cquire (r)elease (k)skip (q)uit [swihwarkq]:
```

03-artifact-acquired.png

P2 locked the Solar Core. The artifact table now reads 'Solar Core: held by P2 (waiters=0)' in yellow, and the log gained an [ACQ] P2 LOCKED Solar Core entry. This is the resource table from \$7 being driven through the shared memory segment under the artifact mutex.

```
OS-proj : arbiter — Konsole
File Edit View Bookmarks Plugins Settings Help
New Tab Split View Copy Paste Find

CHROMO RIFT seed=240673 players=2 enemies=7 killed=0/10

-- PLAYERS --
P1 HP [oooooooooooooooooooo] 240946/240946
  STAMINA [oooooooooooooooooooo] 50/100
P2 HP [oooooooooooooooooooo] 241356/241356
  STAMINA [oooooooooooooooooooo] 0/100

-- ENEMIES --
E1 HP [oooooooooooooooooooo] 169/169
  STAMINA [oooooooooooooooooooo] 50/150
E2 HP [oooooooooooooooooooo] 271/271
  STAMINA [oooooooooooooooooooo] 65/150
E3 HP [oooooooooooooooooooo] 241/241
  STAMINA [oooooooooooooooooooo] 55/150
E4 HP [oooooooooooooooooooo] 264/264
  STAMINA [oooooooooooooooooooo] 70/150
E5 HP [oooooooooooooooooooo] 268/268
  STAMINA [oooooooooooooooooooo] 55/150
E6 HP [oooooooooooooooooooo] 155/155
  STAMINA [oooooooooooooooooooo] 130/150
E7 HP [oooooooooooooooooooo] 176/176
  STAMINA [oooooooooooooooooooo] 100/150

-- ARTIFACTS --
Solar Core : held by P2 (waiters=0)
Lunar Blade : held by P2 (waiters=0)

-- LOG --
Game started, seed=240673 players=2 enemies=7
[asp] 7 NPC threads running
[hup] 2 player thread(s) running
[TURN] P1 begins turn (stamina=100)
[TIMEOUT] P1 did not respond -@-T auto Skip
[SKIP] P1 skipped
[TURN] P2 begins turn (stamina=100)
[ACQ] P2 LOCKED Solar Core
[TURN] P1 begins turn (stamina=100)
[?] P1 submitted unknown action 0
[TURN] P2 begins turn (stamina=100)
[ACQ] P2 LOCKED Lunar Blade

P1 turn: (s)trike (x)exhaust (w)p (l)owp (h)eal (u)lt (a)cquire (r)elease (k)skip (q)uit [swihwarkq]:
```

04-both-artifacts-held.png

P2 now holds both Solar Core AND Lunar Blade. Both entries in the artifact table read 'held by P2'. This is the eligibility precondition for the Ultimate Ability (§8): a player may only trigger Ultimate while both artifacts are simultaneously in their primary inventory.

```
OS-proj : arbiter — Konsole
File Edit View Bookmarks Plugins Settings Help
New Tab Split View
CHROMO RIFT seed=240673 players=2 enemies=7 killed=3/*** ULTIMATE ACTIVE (10s) 6-Q-T ASP SIGSTOPped ***

-- PLAYERS --
P1 HP [*****] 240946/240946
  STAMINA [*****] 50/100
P2 HP [*****] 241339/241356
  STAMINA [*****] 0/100

-- ENEMIES --
E1 HP [*****] -31/169
  STAMINA [*****] 0/150
E2 HP [*****] 71/271
  STAMINA [*****] 91/150
E3 HP [*****] 41/241
  STAMINA [*****] 77/150
E4 HP [*****] 64/264
  STAMINA [*****] 96/150
E5 HP [*****] 68/268
  STAMINA [*****] 77/150
E6 HP [*****] -45/155
  STAMINA [*****] 0/150
E7 HP [*****] -24/176
  STAMINA [*****] 0/150

-- ARTIFACTS --
Solar Core : held by P2 (waiters=0)
Lunar Blade : held by P2 (waiters=0)
Eclipse Relic : FREE (waiters=0)

-- LOG --
Game started, seed=240673 players=2 enemies=7
[asp] 7 NPC threads running
[hip] 2 player thread(s) running
[TURN] P1 begins turn (stamina=100)
[TIMEOUT] P1 did not respond 6-Q-T auto Skip
[SKIP] P1 skipped
[TURN] P2 begins turn (stamina=100)
[ACQ] P2 LOCKED Solar Core
[TURN] P1 begins turn (stamina=100)
[P] P1 submitted unknown action 0
[TURN] P2 begins turn (stamina=100)
[ACQ] P2 LOCKED Lunar Blade
[TURN] P1 begins turn (stamina=100)
[P] P1 submitted unknown action 0
[TURN] E6 begins turn (stamina=150)
[HTT] E6 -> P2 for 17 dmg (hp 241339)
[TURN] P2 begins turn (stamina=100)
[ULT] P2 triggers Ultimate 6-Q-T ASP paused 10s
[ULT] E1 takes 200 dmg
[DEATH] E1 has fallen
[DROP] E1 dropped Splinter Stick 6-Q-T any player may pick it
[ULT] E2 takes 200 dmg
[ULT] E3 takes 200 dmg
[ULT] E4 takes 200 dmg
[ULT] E5 takes 200 dmg
[ULT] E6 takes 200 dmg
[DEATH] E6 has fallen
[ULT] E7 takes 200 dmg
[DEATH] E7 has fallen
[DROP] E7 dropped Venom Dagger 6-Q-T any player may pick it up
[ARTIFACT] Eclipse Relic has appeared!

P1 6-Q-T pick up Venom Dagger? (y/n) [yn]:
```

05-ultimate-banner.png

Ultimate Ability triggered by P2. The magenta banner at the top of the HUD reads '*** ULTIMATE ACTIVE (10s) ... ASP SIGSTOPped ***' — the arbiter sent SIGSTOP to the asp process, armed a 10 s SIGALRM, and dealt 200 dmg to every alive enemy. Log shows five [ULT] rows applying damage, three [DEATH] rows (E1, E6, E7), two [DROP] rows (Splinter Stick, Venom Dagger) and [ARTIFACT] Eclipse Relic has appeared — Eclipse Relic is introduced dynamically at run-time per §7.

```
OS-proj : arbiter — Konsole
File Edit View Bookmarks Plugins Settings Help
New Tab Split View Copy Paste Find

CHROMO RIFT seed=240673 players=2 enemies=7 killed=4/10

-- PLAYERS --
P1 HP [oooooooooooooooooooo] 240946/240946
  STAMINA [oooooooooooooooooooo] 100/100
P2 HP [oooooooooooooooooooo] 241339/241356
  STAMINA [oooooooooooooooooooo] 50/100

-- ENEMIES --
E1 HP [oooooooooooooooooooo] -31/169
  STAMINA [oooooooooooooooooooo] 0/150
E2 HP [oooooooooooooooooooo] -24/271
  STAMINA [oooooooooooooooooooo] 0/150
E3 HP [oooooooooooooooooooo] 41/241
  STAMINA [oooooooooooooooooooo] 132/150
E4 HP [oooooooooooooooooooo] 64/264
  STAMINA [oooooooooooooooooooo] 0/150
E5 HP [oooooooooooooooooooo] 68/268
  STAMINA [oooooooooooooooooooo] 132/150
E6 HP [oooooooooooooooooooo] -45/155
  STAMINA [oooooooooooooooooooo] 0/150
E7 HP [oooooooooooooooooooo] -24/176
  STAMINA [oooooooooooooooooooo] 0/150

-- ARTIFACTS --
Solar Core : held by P2 (waiters=0)
Lunar Blade : held by P2 (waiters=0)
Eclipse Relic : held by E4 (waiters=0)

-- LOG --
Game started, seed=240673 players=2 enemies=7
[asp] 7 NPC threads running
[hlp] 2 player thread(s) running
[TURN] P1 begins turn (stamina=100)
[TIMEOUT] P1 did not respond <->-T auto Skip
[SKIP] P1 skipped
[TURN] P2 begins turn (stamina=100)
[ACQ] P2 LOCKED Solar Core
[TURN] P1 begins turn (stamina=100)
[P?] P1 submitted unknown action 0
[TURN] P2 begins turn (stamina=100)
[ACQ] P2 LOCKED Lunar Blade
[TURN] P1 begins turn (stamina=100)
[P?] P1 submitted unknown action 0
[TURN] E6 begins turn (stamina=150)
[HIT] E6 -> P2 for 17 dmg (hp 241339)
[TURN] P2 begins turn (stamina=100)
[ULT] P2 triggers Ultimate <->-T ASP paused 10s
[ULT] E1 takes 200 dmg
[DEATH] E1 has fallen
[DROP] E1 dropped Salienter Stick <->-T any player may pick it
[ULT] E2 takes 200 dmg
[ULT] E3 takes 200 dmg
[ULT] E4 takes 200 dmg
[ULT] E5 takes 200 dmg
[ULT] E6 takes 200 dmg
[ULT] E7 takes 200 dmg
[DEATH] E6 has fallen
[DEATH] E7 has fallen
[DROP] E7 dropped Venom Dagger <->-T any player may pick it up
[ARTIFACT] Eclipse Relic has appeared!
[TURN] P1 begins turn (stamina=100)
[P?] P1 submitted unknown action 0
[TURN] P2 begins turn (stamina=100)
[ULT] Ultimate pause ended <->-T ASP resumed
[DROP] Venom Dagger expired <->-T E3 grabbed Venom Dagger
[TURN] P1 begins turn (stamina=100)
[P?] P1 submitted unknown action 0
[TURN] P2 begins turn (stamina=100)
[HIT] P2 -> E2 for 95 dmg (hp -24)
[DEATH] E2 has fallen
[DROP] E2 dropped Thunderstaff <->-T any player may pick it up
[USE] P2 wields Solar Core (95 dmg)
[TURN] E4 begins turn (stamina=150)
[ACQ] E4 LOCKED Eclipse Relic

P1 <->-T pick up Thunderstaff? (y/n) [yn]:
```

06-weapon-use-kill-drop.png

P2 uses the Solar Core as a weapon. The log shows [USE] P2 wields Solar Core (95 dmg), [HIT] P2 -> E2 for 95 dmg, [DEATH] E2 has fallen, [DROP] E2 dropped Thunderstaff. Immediately after, [ACQ] E4 LOCKED Eclipse Relic — an NPC grabbed the Relic, demonstrating that the asp threads contend for artifacts symmetrically with players. The artifact table now shows 'Eclipse Relic: held by E4' in yellow, which would enter the waits-for graph if any player subsequently acquired it (\$7 deadlock detector).

Analysis

The scheduler honoured the stamina-arrival rule from spec §3: entities only committed actions once their stamina reached `max_stamina`, and the scheduler picked the earliest-ready entity (FIFO by `last_full_at_ns`, not by category). Over this demo run we observed:

- Total player turns : 80
- Total enemy turns : 3
- Enemies killed : 7/10
- Avg player wait : 100.2 ms
- Avg enemy wait : 400.5 ms

Players have higher speed (`100/num_players` vs 10–30 for enemies) so they fill their stamina gauge faster and therefore act more often – matching the intended asymmetry of the game.

The Gantt chart shows the strict serial-execution property: no two action bars ever overlap in time, consistent with '§3 Serial Execution – only one character may perform an action at any given moment'.

Any 3-second burst segments in the enemy rows correspond to the NPC-turn-timeout auto-skip (spec §8). These occur when the ASP is paused by an active Ultimate (SIGSTOP) and the arbiter's 3s `sem_timedwait` elapses before an action is committed – the arbiter then records a Skip, as required.

Stamina accrual is enforced at 1 Hz via a `last_tick_ns` guard inside `scheduler_tick()`; the scheduler loop itself wakes more often (every 200 ms) only so stun/Ultimate/shutdown events are observed promptly. This decoupling keeps the spec's 'Each second the entity's speed is added to its current stamina' invariant intact without blocking the action-commit path.